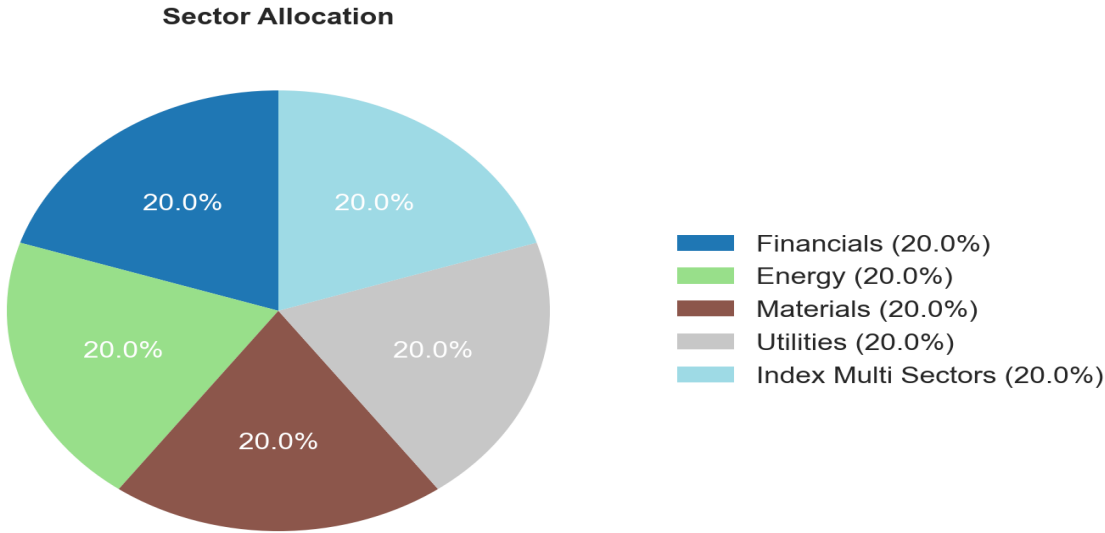


Portfolio rotation report

Date of extraction	2025-12-07 21:41
Data window	2021-01-01 → 2022-12-01
Frequency	Monthly — 24 observations. Sufficient depth: short- and long-term indicators are statistically reliable.
Tickers	SPY, XLB, XLE, XLF, XLU
Indicators	Short term: 3 months volatility, 4 months return, 7 months momentum Long term: 7 months volatility, 8 months return, 21 months momentum

1. Portfolio Composition

- XLF (Financial Select Sector SPDR Fund — Financials)
- XLE (Energy Select Sector SPDR Fund — Energy)
- XLB (Materials Select Sector SPDR Fund — Materials)
- XLU (Utilities Select Sector SPDR Fund — Utilities)
- SPY (SPDR S&P; 500 ETF — Index Multi Sectors)



Macro Profile Insight

Macro coverage

- Partial coverage: several macro themes remain unrepresented.

Sectorial biases

- Inflation-sensitive bias: protection against rising prices but higher volatility.

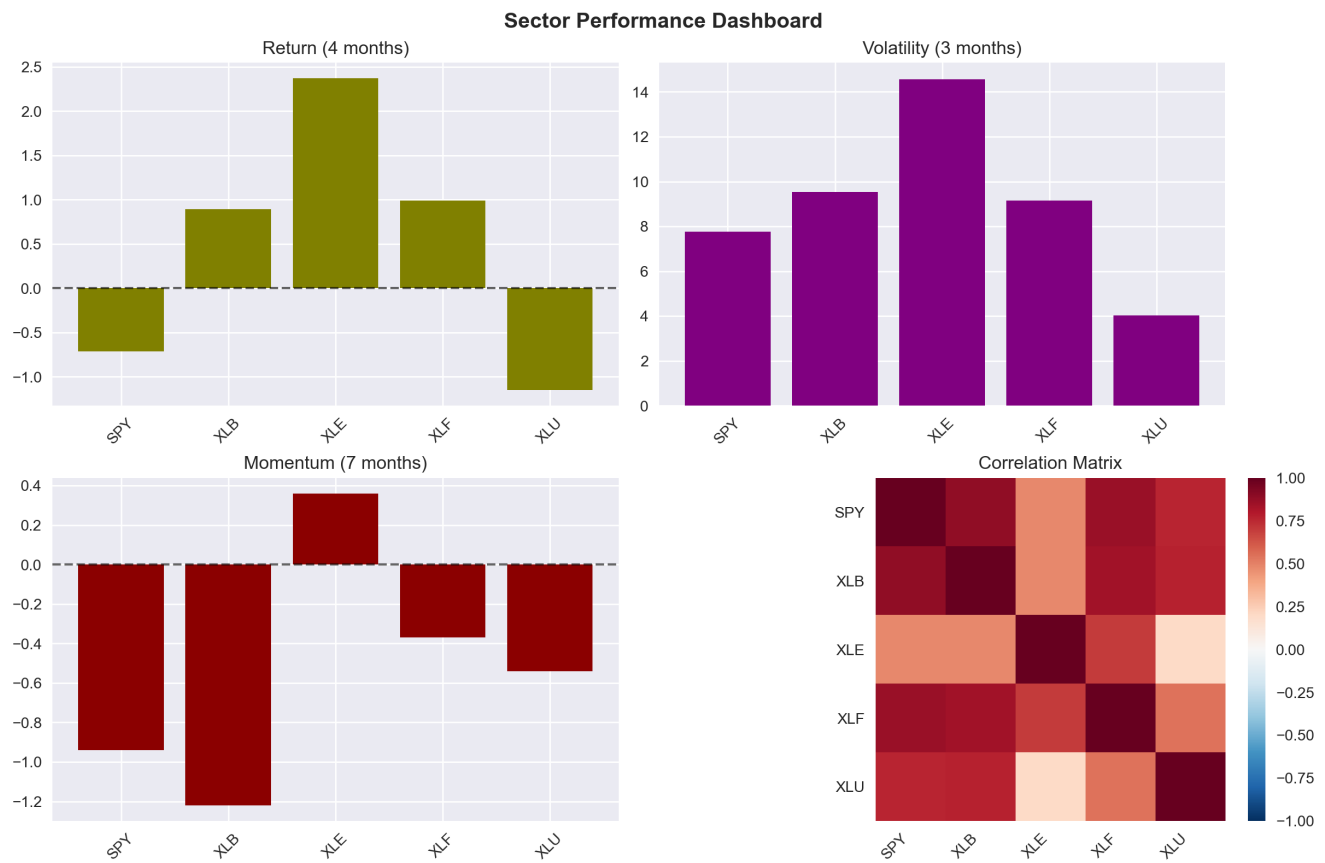
Structural properties

- Mid-cap tilt: balanced exposure combining quality and residual growth convexity.
- Concentration risk: portfolio heavily exposed to etf instruments.
- Liquidity stance: pockets of thin/illiquid names; robust median dollar volume.
- Market-neutral beta exposure: close alignment with index dynamics.
- Balanced volatility structure: risk evenly distributed across holdings.

Additional comments

- Significant sector gaps: broad portions of the economy remain uncovered.

2. Sector Performance Visualization



Top Positive Correlations

90.0% of portfolio

Pair	Correlation
SPY / XLB	0.89
SPY / XLF	0.86
XLB / XLF	0.84
XLB / XLU	0.77
SPY / XLU	0.76

Top Neutral Correlations

10.0% of portfolio

Pair	Correlation
XLE / XLU	0.2

Overall, average cross-sector correlation was 0.65, ranging from 0.20 to 0.89. This suggests very high correlation — sectors behave almost as a single macro block. Correlation dispersion remains meaningful, indicating both tightly and weakly linked sectors. Strongest co-movement between SPY and XLB (0.89); weakest between XLE and XLU (0.20). Market-driven regime: macro factors dominate individual sector dynamics.

3. Key Performance Indicators

Ticker	R (%)	V (%)	RAR	M (%)	Signal	Regime	Liquidity	Commentary
SPY	-0.71	7.76	-0.092	-0.94	neutral	subdued	liquid	weak performance with negative momentum, signal unclear or in transition, low volatility — calm market tone, with ample liquidity.
XLB	0.89	9.53	0.093	-1.22	neutral	normal	thin	positive returns but fading momentum, signal unclear or in transition, normal volatility regime, under limited liquidity.
XLE	2.37	14.57	0.163	0.36	mildly coherent	subdued	liquid	mixed short-term dynamics, signal remains coherent, but with limited conviction, low volatility — calm market tone, with ample liquidity.
XLF	0.99	9.15	0.108	-0.37	neutral	subdued	thin	positive returns but fading momentum, signal unclear or in transition, low volatility — calm market tone, under limited liquidity.
XLU	-1.15	4.02	-0.286	-0.54	weak reversal	normal	thin	weak performance with negative momentum, early reversal signs, short-term momentum starting to diverge, normal volatility regime, under limited liquidity.

Top by Average Return

Ticker	AvgReturn(%)
XLE	2.37
XLF	0.99
XLB	0.89
SPY	-0.71
XLU	-1.15

Top by Volatility

Ticker	Volatility(%)
XLE	14.57
XLB	9.53
XLF	9.15
SPY	7.76
XLU	4.02

Top by Momentum

Ticker	Momentum(%)
XLE	0.36
XLF	-0.37
XLU	-0.54
SPY	-0.94
XLB	-1.22

3.b Rotation & Stress Metrics

Ticker	ΔReturn (pp)	Momentum Ratio	Signal	Volatility Ratio	Beta	ADV10 USD	Interpretation
SPY	0.38	9.40	0.09	0.93	1.00	72.91 M	stable performance, momentum surging, volatility normalizing, within defensive profile, under thin liquidity regime.
XLB	2.28	-12.20	-0.092	0.92	0.82	11.78 M	acceleration in returns, momentum fading, volatility normalizing, within defensive profile, under thin liquidity regime.
XLE	1.06	0.08	0.333	1.03	0.80	28.61 M	acceleration in returns, momentum fading, volatility normalizing, within defensive profile, under thin liquidity regime.
XLF	1.29	-2.10	-0.119	1.01	0.84	36.84 M	acceleration in returns, momentum fading, volatility normalizing, within defensive profile, under thin liquidity regime.
XLU	-1.00	-0.71	-0.316	0.59	0.42	19.35 M	stable performance, momentum fading, volatility compressing, within defensive profile, under thin liquidity regime.

■ Light-impact observations

For **SPY, XLE, XLF**: *Volatility compression* — subdued volatility; a breakout setup may be forming. (avg ann. return 0.9%, momentum -0.3%, vol 10.5%)

For **XLB, XLU**: *Range / noise* — no clear direction; rotating momentum and mid-range volatility. (avg ann. return -0.1%, momentum -0.9%, vol 6.8%)